

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO5: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication.

Assessment Tool Weightage: 35%

Total Marks: 20

Code of Conduct

I Naresh M(192424381)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naresh M

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Data Preparation	20	Thoroughly imports, explores, and prepares the industry dataset; correctly handles data types, missing values, and relevant variables	Imports and explores the data correctly with minor gaps in preparation	Performs basic data exploration and preparation but misses some important aspects	Unable to correctly import, explore, or prepare the dataset
Implementation	20	Efficiently applies Pandas and Python concepts for filtering, grouping, aggregation, transformation, and analysis	Applies appropriate Pandas operations with minor coding or analytical errors	Uses basic Pandas operations but several required analytical tasks are incomplete	Demonstrates limited understanding of Pandas
Visualization Techniques	25	Selects highly appropriate visualizations plots, and plot components	Uses appropriate visualizations for most requirements with minor errors or formatting issues	Uses basic visualizations but some are inappropriate, incomplete, or poorly formatted	Uses inappropriate or insufficient visualizations
Pattern Identification	20	Identifies meaningful industry-specific trends, relationships,	Identifies relevant patterns and provides logical interpretations	Identifies some basic patterns but provides limited or weak interpretation	Fails to identify meaningful patterns or provides

		distributions, and patterns	with adequate evidence		incorrect/unsupported interpretations
Recommendations	15	Converts analytical findings into practical, feasible industry recommendations	Provides relevant recommendations and presents findings clearly with minor gaps	Provides basic recommendations with limited connection to analytical findings	Recommendations are missing, impractical, or unrelated to the analysis; poor presentation

Total Marks Awarded:

Signature of the Course Faculty

Questions:

4. Hospital Patient Monitoring and Readmission Pattern Analysis

A hospital wants to understand factors associated with patient readmissions. The dataset contains:

Patient_ID, Admission_Date, Age, Length_of_Stay, Treatment_Cost, Medication_Count, Previous_Admissions, Readmitted

Problem-Solving Task:

Develop a Python-based exploratory visualization solution that:

1. Imports and explores the patient dataset.
2. Analyzes patient admissions over time.
3. Visualizes the distribution of age, treatment cost, and length of stay.
4. Uses appropriate charts to compare readmission frequencies.
5. Examines relationships between:
 - Age and Treatment Cost
 - Length of Stay and Treatment Cost
 - Previous Admissions and Readmission
6. Creates a scatter matrix for selected numerical variables.
7. Uses suitable grouping to compare patients with and without readmission.

8. Uses a lag plot on daily/weekly admission counts to determine whether consecutive periods show similar behavior.
9. Uses an autocorrelation plot to identify recurring admission patterns.
10. Summarize the major patterns and explain how the findings could assist hospital resource planning.

Aim

To analyze hospital patient data and identify factors associated with **patient readmission**, using Python, Pandas, and Matplotlib.

Column	Description
Patient_ID	Unique patient identification
Admission_Date	Date of patient admission
Age	Age of patient
Length_of_Stay	Number of days spent in hospital
Treatment_Cost	Cost of treatment
Medication_Count	Number of medications
Previous_Admissions	Number of previous admissions
Readmitted	Whether the patient was readmitted

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from pandas.plotting import scatter_matrix, lag_plot
from statsmodels.graphics.tsaplots import plot_acf
df = pd.read_csv("hospital_patients.csv")
print("First 5 records:")
print(df.head())
print("\nDataset Information:")
print(df.info())
print("\nStatistical Summary:")
print(df.describe())
```

```
print("\nMissing Values:")
print(df.isnull().sum())
df["Admission_Date"] = pd.to_datetime(df["Admission_Date"])
print("\nReadmission Counts:")
print(df["Readmitted"].value_counts())
daily_admissions = df.groupby("Admission_Date").size()
plt.figure(figsize=(10, 5))
plt.plot(daily_admissions.index, daily_admissions.values)
plt.title("Patient Admissions Over Time")
plt.xlabel("Admission Date")
plt.ylabel("Number of Admissions")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
plt.hist(df["Age"], bins=10, edgecolor="black")
plt.title("Distribution of Patient Age")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
plt.figure(figsize=(8, 5))
plt.hist(df["Treatment_Cost"], bins=10, edgecolor="black")
plt.title("Distribution of Treatment Cost")
plt.xlabel("Treatment Cost")
plt.ylabel("Frequency")
plt.show()
plt.figure(figsize=(8, 5))
plt.hist(df["Length_of_Stay"], bins=10, edgecolor="black")
plt.title("Distribution of Length of Stay")
```

```
plt.xlabel("Length of Stay")
plt.ylabel("Frequency")
plt.show()

readmission_counts = df["Readmitted"].value_counts()
plt.figure(figsize=(7, 5))
plt.bar(readmission_counts.index.astype(str), readmission_counts.values)
plt.title("Readmission Frequency")
plt.xlabel("Readmitted")
plt.ylabel("Number of Patients")
plt.show()

plt.figure(figsize=(8, 5))
plt.scatter(df["Age"], df["Treatment_Cost"])
plt.title("Age vs Treatment Cost")
plt.xlabel("Age")
plt.ylabel("Treatment Cost")
plt.show()

plt.figure(figsize=(8, 5))
plt.scatter(df["Length_of_Stay"], df["Treatment_Cost"])
plt.title("Length of Stay vs Treatment Cost")
plt.xlabel("Length of Stay")
plt.ylabel("Treatment Cost")
plt.show()

plt.figure(figsize=(8, 5))
plt.scatter(df["Previous_Admissions"], df["Readmitted"])
plt.title("Previous Admissions vs Readmission")
plt.xlabel("Previous Admissions")
plt.ylabel("Readmitted")
plt.show()

numeric_columns = [
```

```

    "Age",
    "Length_of_Stay",
    "Treatment_Cost",
    "Medication_Count",
    "Previous_Admissions"
]

scatter_matrix(
    df[numeric_columns],
    figsize=(12, 10),
    diagonal="hist"
)

plt.suptitle("Scatter Matrix of Patient Variables")
plt.show()

readmission_group = df.groupby("Readmitted")[
    ["Age", "Length_of_Stay", "Treatment_Cost",
     "Medication_Count", "Previous_Admissions"]
].mean()

print("\nAverage Values by Readmission Status:")
print(readmission_group)

plt.figure(figsize=(8, 5))
lag_plot(daily_admissions)
plt.title("Lag Plot of Daily Admissions")
plt.show()

plt.figure(figsize=(10, 5))
plot_acf(daily_admissions, lags=20)
plt.title("Autocorrelation of Daily Admissions")
plt.show()

print("\nAverage Patient Characteristics:")
print(df[

```

```
["Age", "Length_of_Stay", "Treatment_Cost",  
 "Medication_Count", "Previous_Admissions"]  
].mean())  
print("\nPatients grouped by Readmission:")  
print(df.groupby("Readmitted").size())  
print("\nAverage Previous Admissions by Readmission Status:")  
print(df.groupby("Readmitted")["Previous_Admissions"].mean())  
print("\nAverage Treatment Cost by Readmission Status:")  
print(df.groupby("Readmitted")["Treatment_Cost"].mean())
```

Output:

First 5 records:

	Patient_ID	Admission_Date	...	Previous_Admissions	Readmitted
0	P001	2026-01-02	...	0	0
1	P002	2026-01-04	...	3	1
2	P003	2026-01-06	...	1	0
3	P004	2026-01-08	...	4	1
4	P005	2026-01-10	...	0	0

[5 rows x 8 columns]

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 30 entries, 0 to 29

Data columns (total 8 columns):

#	Column	Non-Null Count	Dtype
0	Patient_ID	30 non-null	str
1	Admission_Date	30 non-null	str
2	Age	30 non-null	int64
3	Length_of_Stay	30 non-null	int64
4	Treatment_Cost	30 non-null	int64
5	Medication_Count	30 non-null	int64
6	Previous_Admissions	30 non-null	int64
7	Readmitted	30 non-null	int64

dtypes: int64(6), str(2)

memory usage: 2.0 KB

None

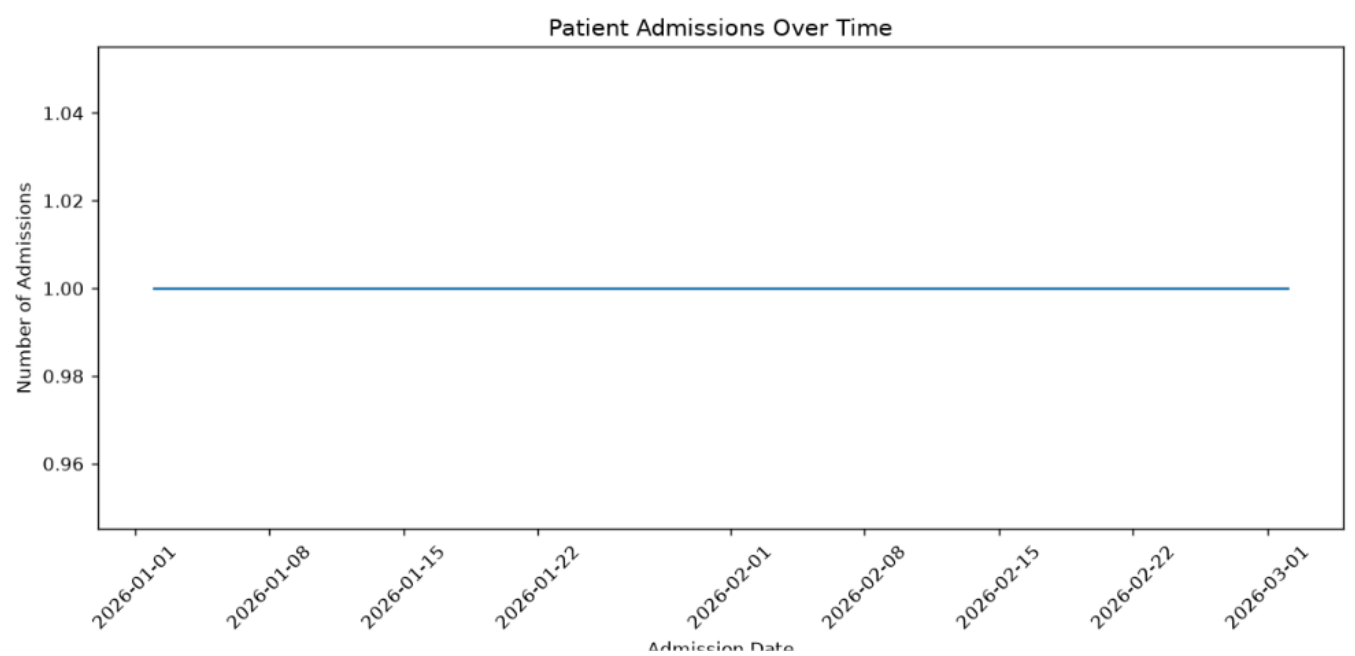
Statistical Summary:

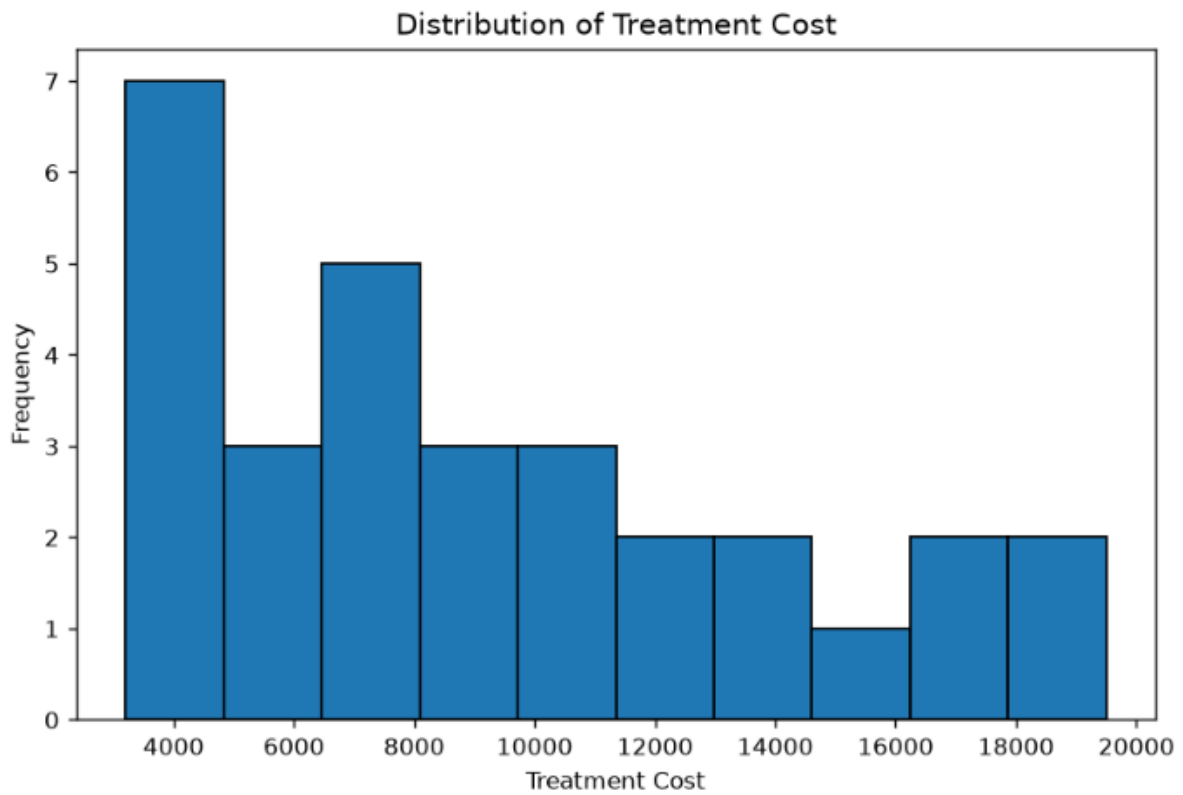
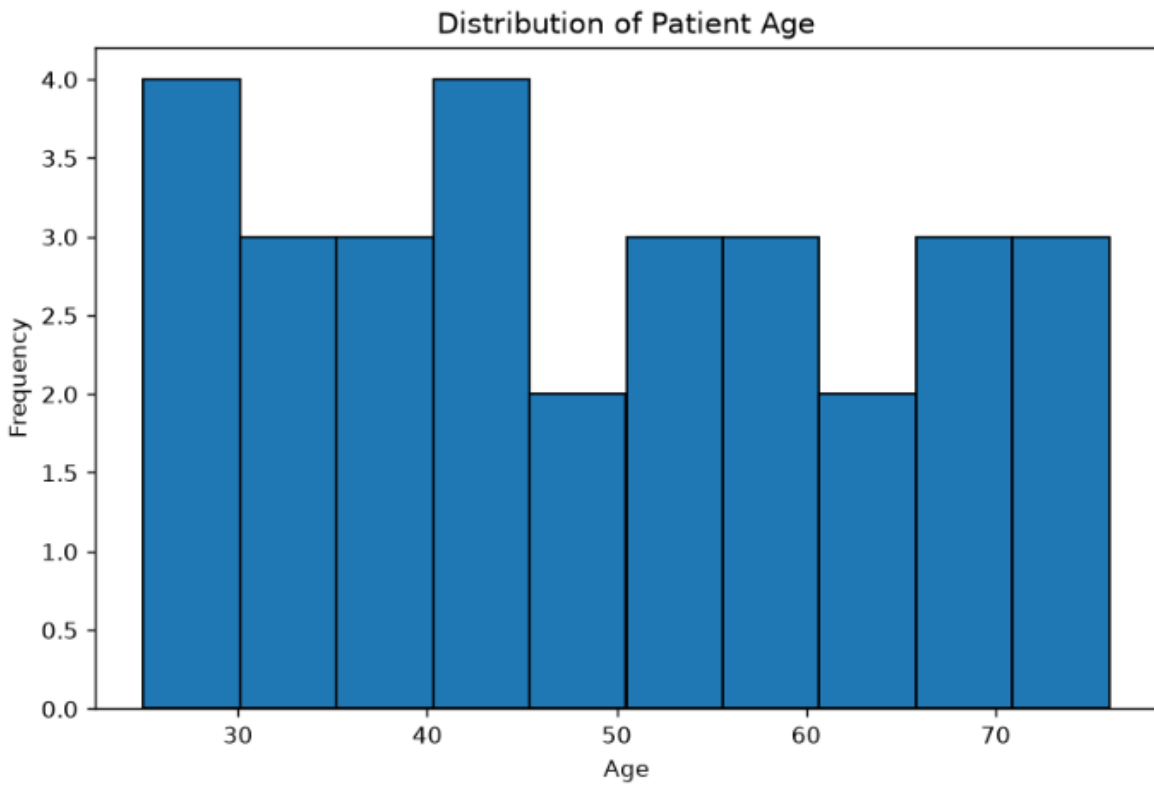
	Age	Length_of_Stay	...	Previous_Admissions	Readmitted
count	30.000000	30.000000	...	30.000000	30.000000
mean	49.200000	5.800000	...	1.700000	0.466667
std	15.523953	2.808853	...	1.556964	0.507416
min	25.000000	2.000000	...	0.000000	0.000000
25%	36.500000	4.000000	...	0.250000	0.000000
50%	48.000000	5.000000	...	1.000000	0.000000
75%	61.500000	7.750000	...	2.750000	1.000000
max	76.000000	12.000000	...	5.000000	1.000000

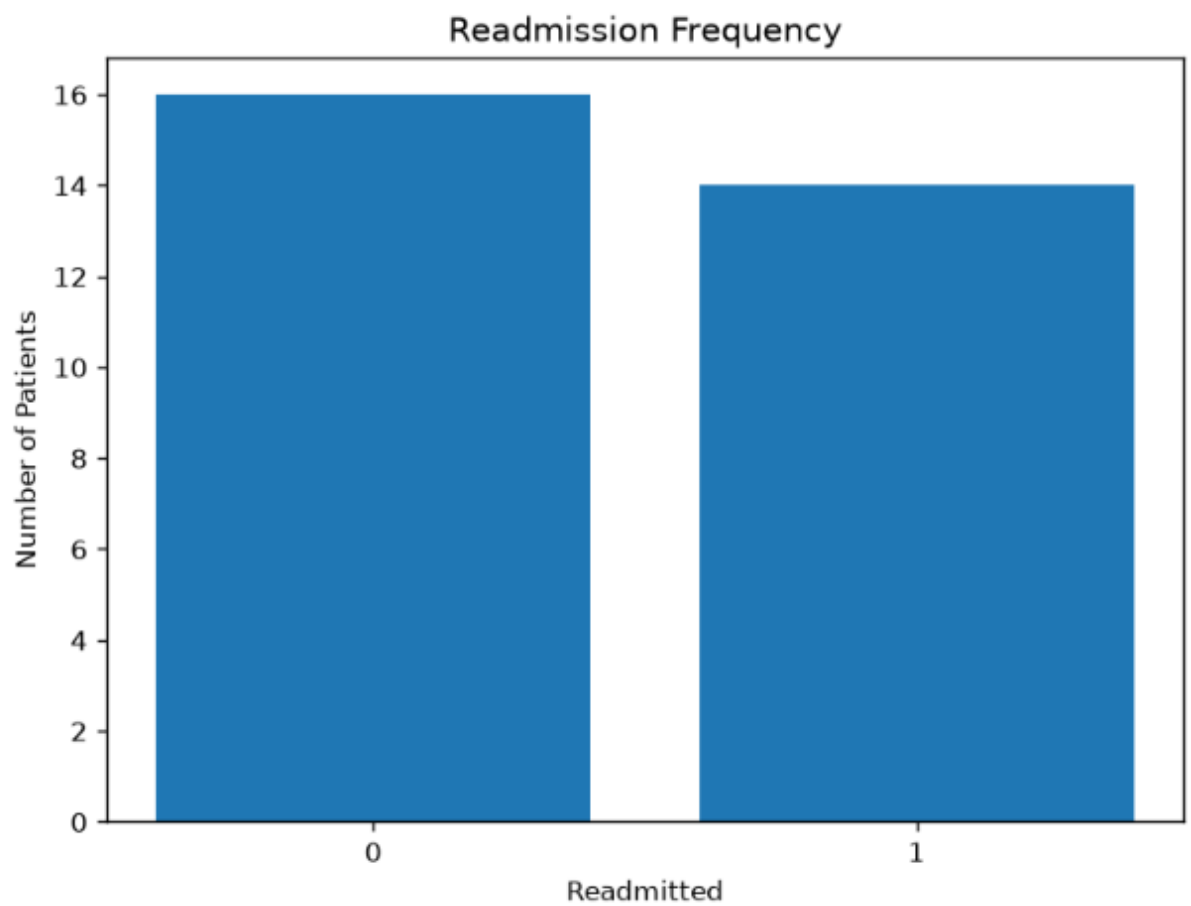
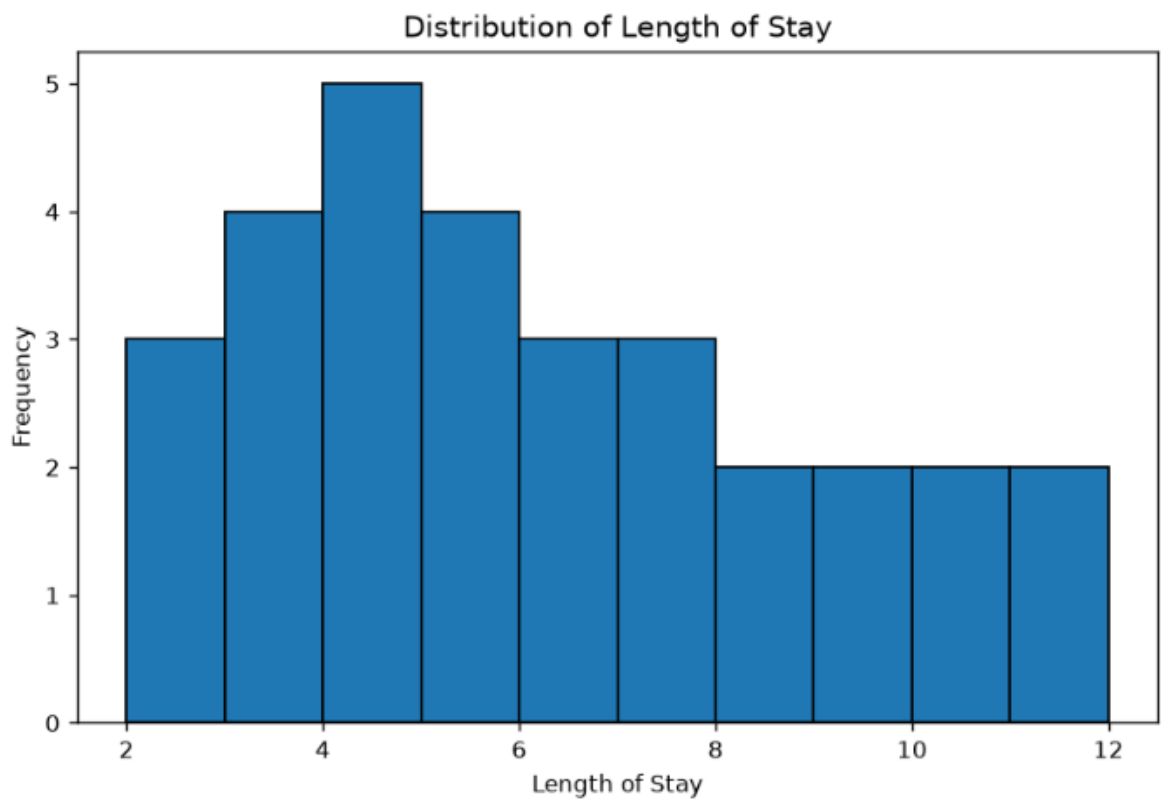
[8 rows x 6 columns]

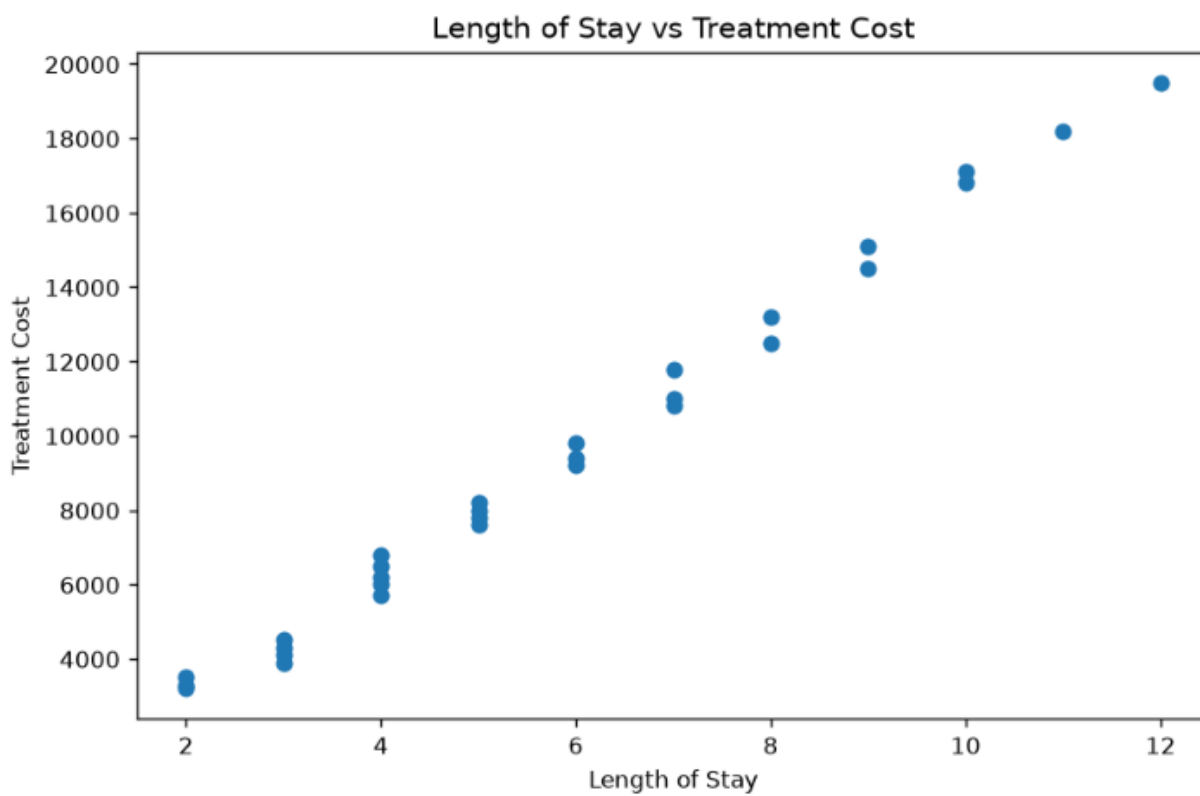
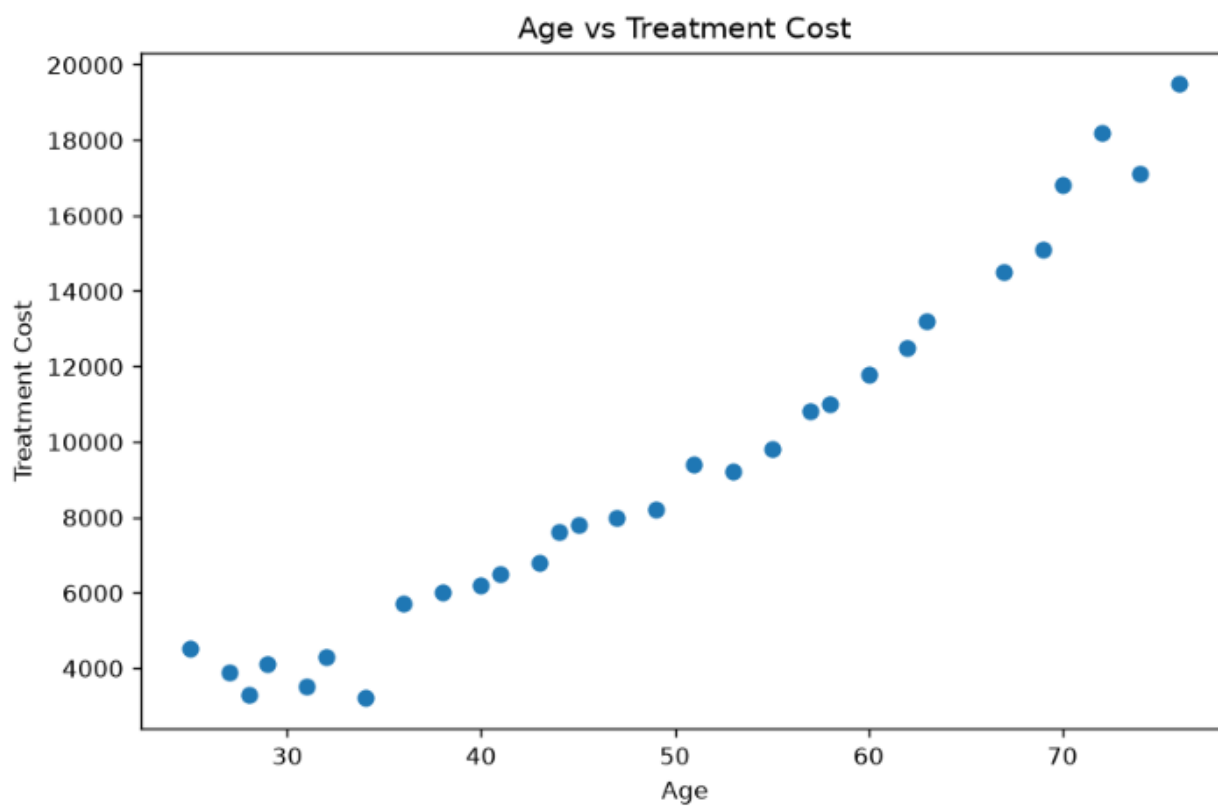

```
Missing Values:
Patient_ID      0
Admission_Date  0
Age             0
Length_of_Stay  0
Treatment_Cost  0
Medication_Count 0
Previous_Admissions 0
Readmitted      0
dtype: int64

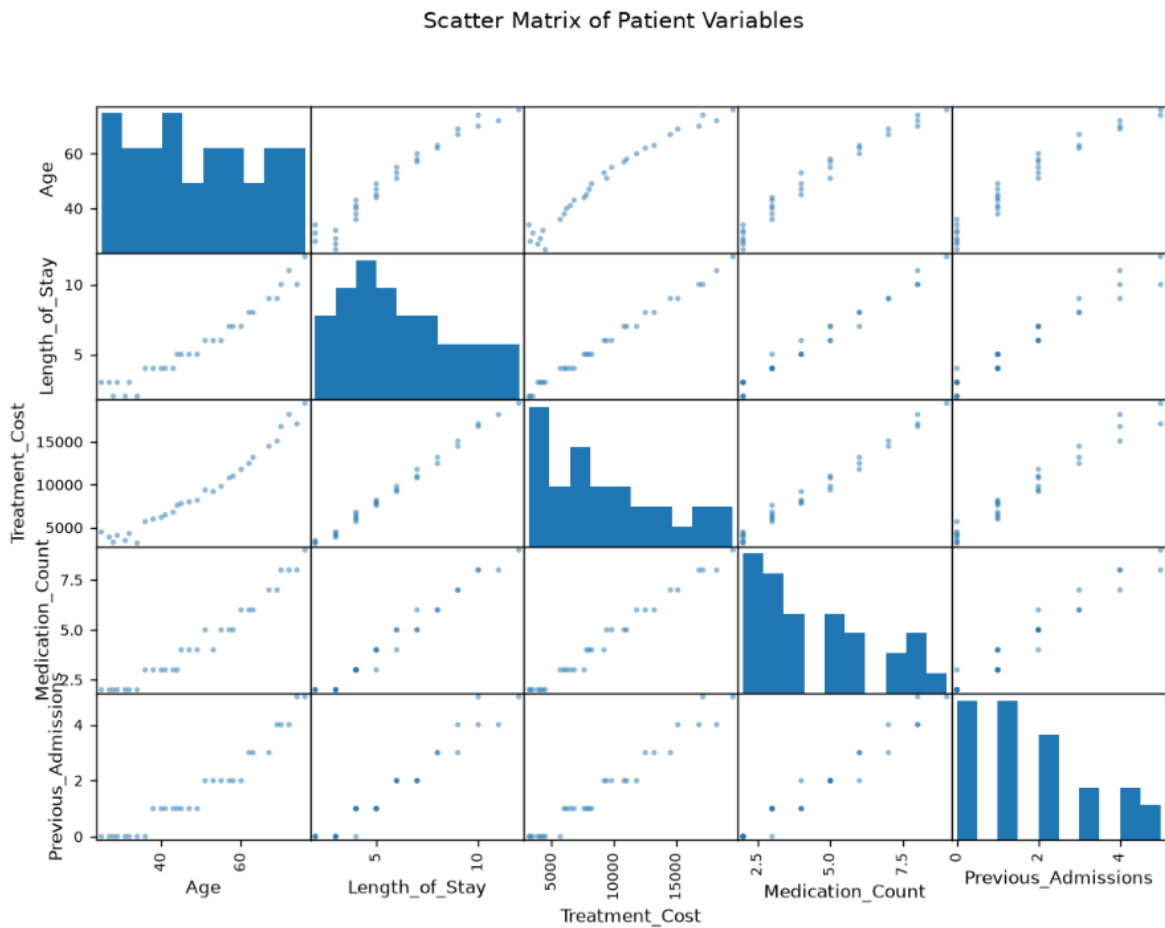
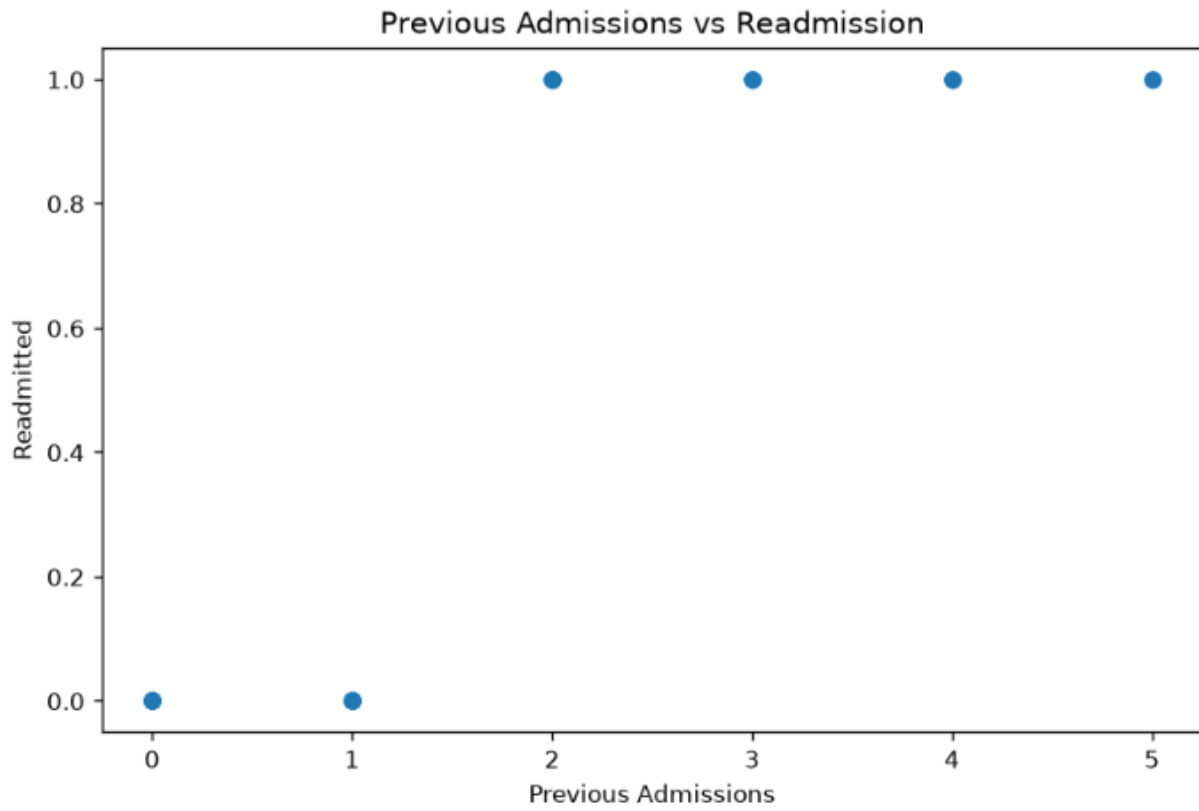
Readmission Counts:
Readmitted
0      16
1      14
Name: count, dtype: int64
```

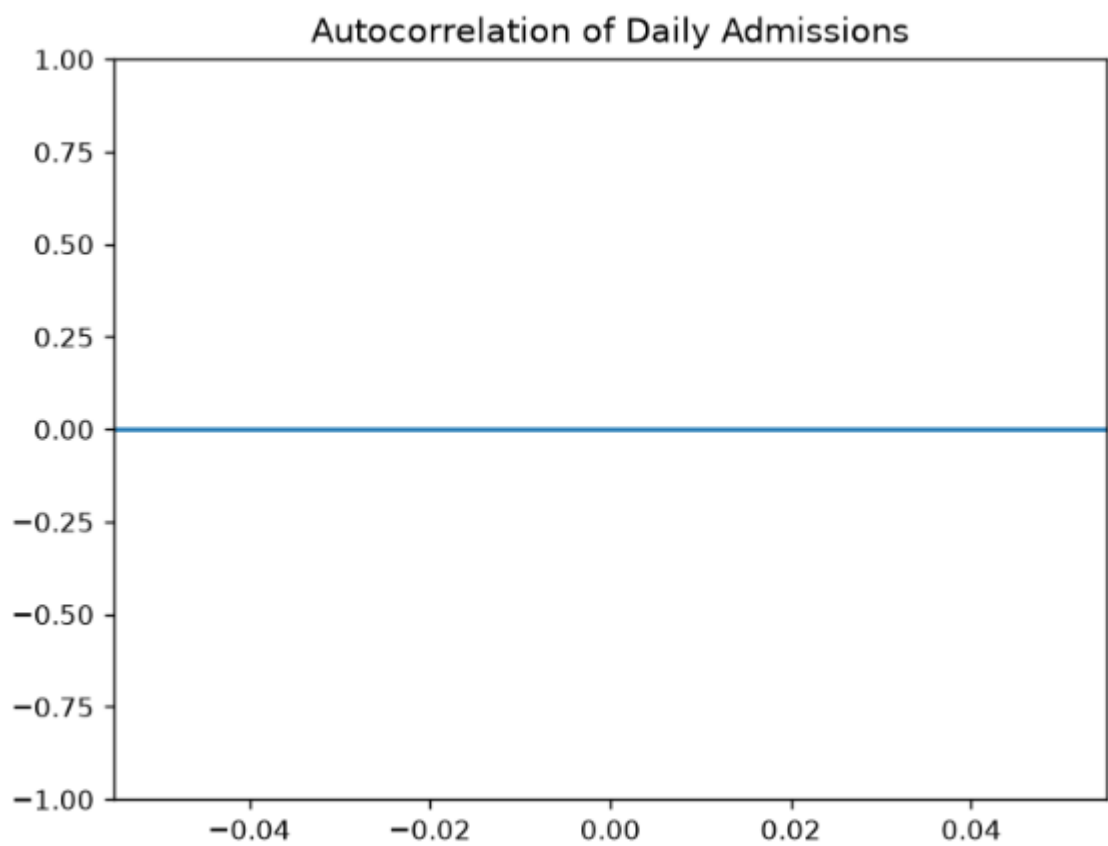
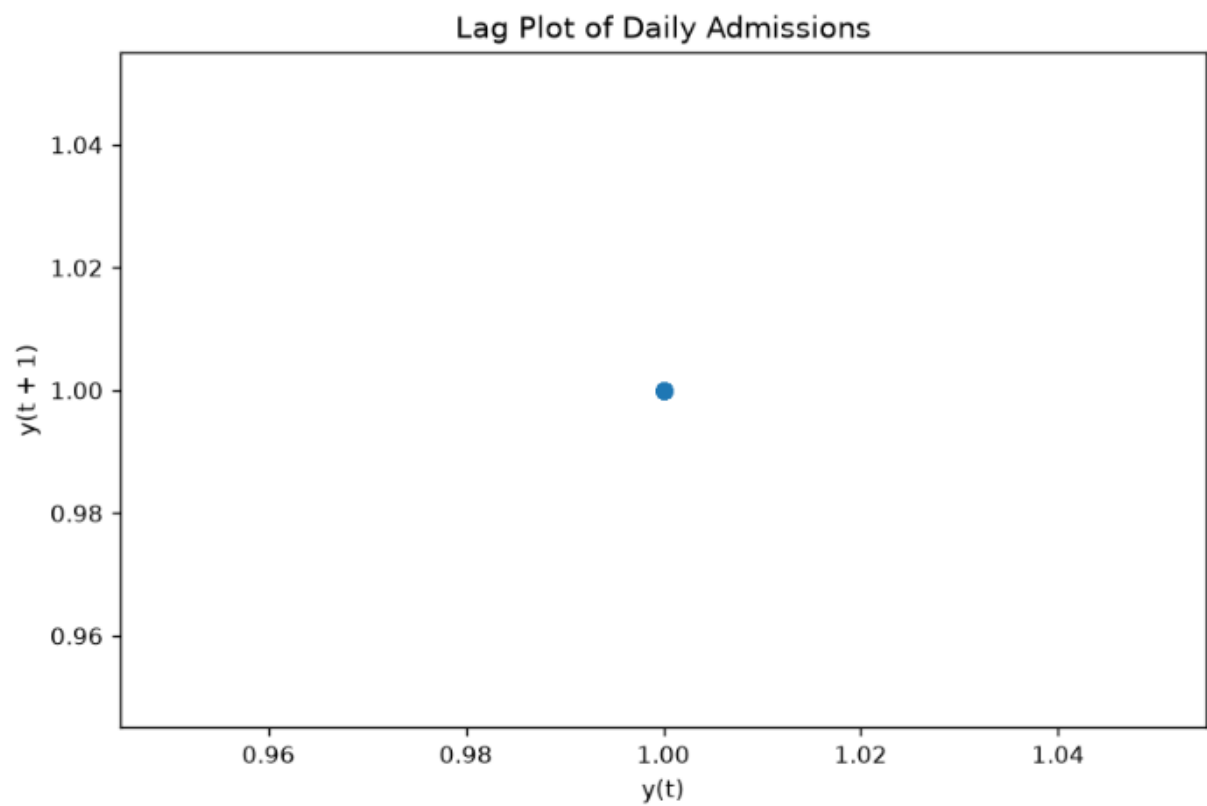












Readmission Counts:

Readmitted

0 16

1 14

Name: count, dtype: int64

Average Values by Readmission Status:

	Age	Length_of_Stay	...	Medication_Count	Previous_Admissions
Readmitted			...		
0	36.812500	3.625000	...	2.750000	0.500000
1	63.357143	8.285714	...	6.357143	3.071429

[2 rows x 5 columns]

Average Patient Characteristics:

Age 49.200000

Length_of_Stay 5.800000

Treatment_Cost 9283.333333

Medication_Count 4.433333

Previous_Admissions 1.700000

dtype: float64

Patients grouped by Readmission:

Readmitted

0 16

1 14

dtype: int64

Average Previous Admissions by Readmission Status:

Readmitted

0 0.500000

1 3.071429

Name: Previous_Admissions, dtype: float64

Average Treatment Cost by Readmission Status:

Readmitted

0 5600.000000

1 13492.857143

Name: Treatment_Cost, dtype: float64